

Part K: Manuscript Consistency Check
Designing Verification Checkpoints to Mitigate Silent Errors in Agentic AI Pipelines

Name: Jahanvi Mishra | Prompt used: “Review this manuscript for consistency in terminology, abbreviations, variable names, mathematical notation, dataset names, model names, capitalization, units, figure references, table references, and citation style. Report inconsistencies only. Do not rewrite the manuscript.”

The full compiled main.tex file was reviewed for internal consistency in wording, notation, and naming. The document was searched systematically for repeated terms, symbols, and labels used in more than one form. Three genuine inconsistencies were identified and are reported below without being corrected, as instructed by the prompt above.

Table: Inconsistencies Found

#	Inconsistency Found	Original Usage	Corrected Usage
1	The symbol used for a general pipeline/system state is not consistent. Some sections use S_i (indexed by checkpoint or step), while other sections use S_t (indexed by time).	$C_i(S_i)$ in Section 4.1; $S_0, S_1, \dots, S_n$ in Section 6; but $I(S_t)$ in Section 5.1 and “For state S_t ” in Section 4.2.	Standardize on a single index variable for state (e.g. always S_i), or explicitly state early in the paper that i and t are used interchangeably to index pipeline steps.
2	The phrase used for the overall system being verified alternates between two forms that refer to the same concept.	“agentic pipeline” (Sections 3.2 and 4.1) versus “agentic AI pipeline” (Section 1 and the Figure 2 caption).	Pick one form (“agentic AI pipeline” is more precise and matches the paper’s title) and use it consistently throughout.
3	Two different checkpoint taxonomies in the paper are named using two different styles, which could be misread as the same five/six-item list.	Section 4.2 names its five checkpoints with ordinal words: “first layer,” “second layer,” etc. Section 8 names its six checkpoints with a numbered label style instead: “Checkpoint 1,” “Checkpoint 2,” etc. (as shown in Figure 2).	Use one consistent naming style for checkpoint layers across both taxonomies, or add a short clarifying sentence noting that the five-layer architecture (Section 4.2) and the six-layer VCP-AI architecture (Section 8) are related but distinct frameworks.

Note

Items 1 and 3 above reflect notation and structure already present in the original research paper (the S_i / S_t mismatch and the two separate checkpoint taxonomies both exist in the source document). They were preserved as-is during conversion rather than silently corrected, in keeping with the rule that AI-assisted conversion should not reinterpret or alter the original author’s scientific content. Item 2 is a wording choice made during paraphrasing and could be standardized without affecting scientific meaning.